Criterion B: Solution overview

Flow Charts

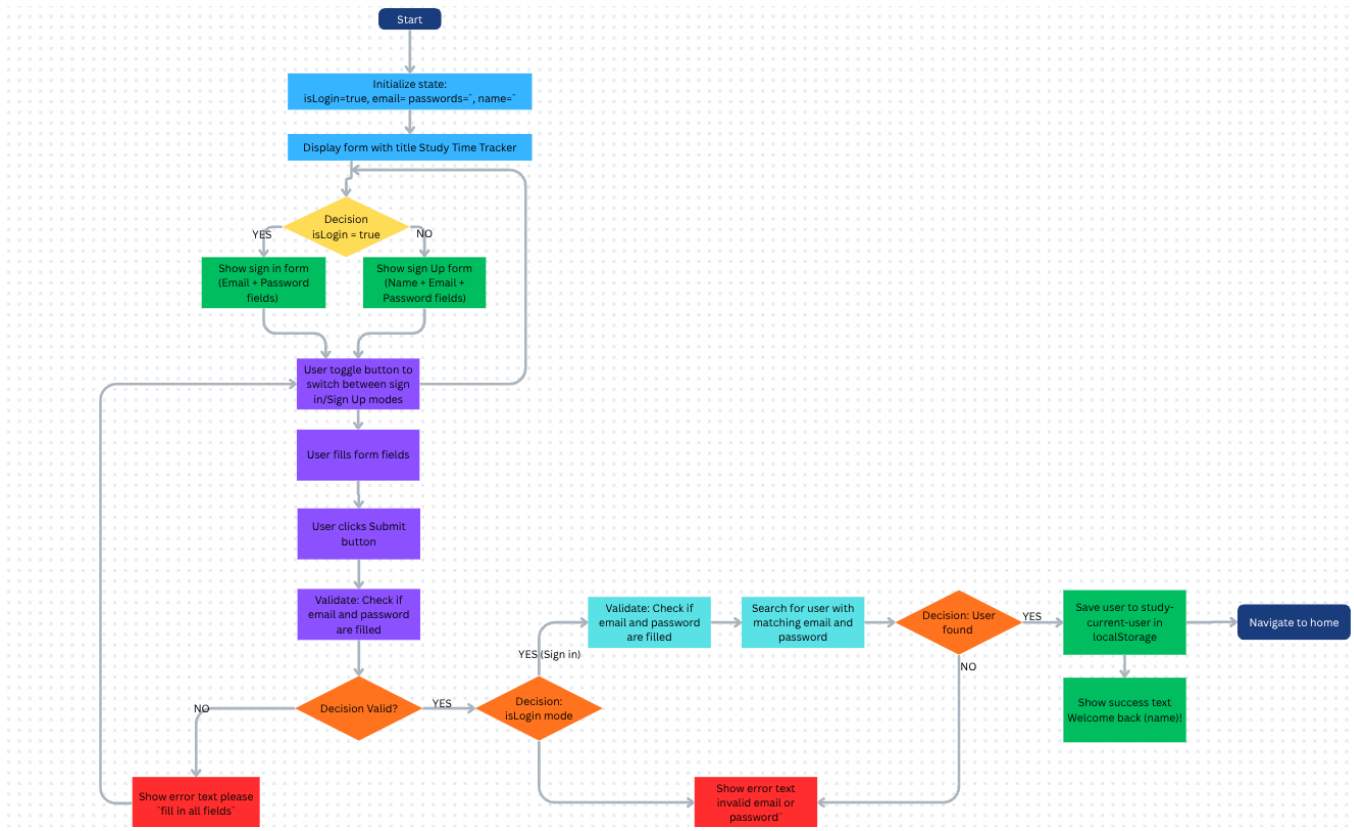

Figure 1: Flowchart for the authentication process of implementing the username and password

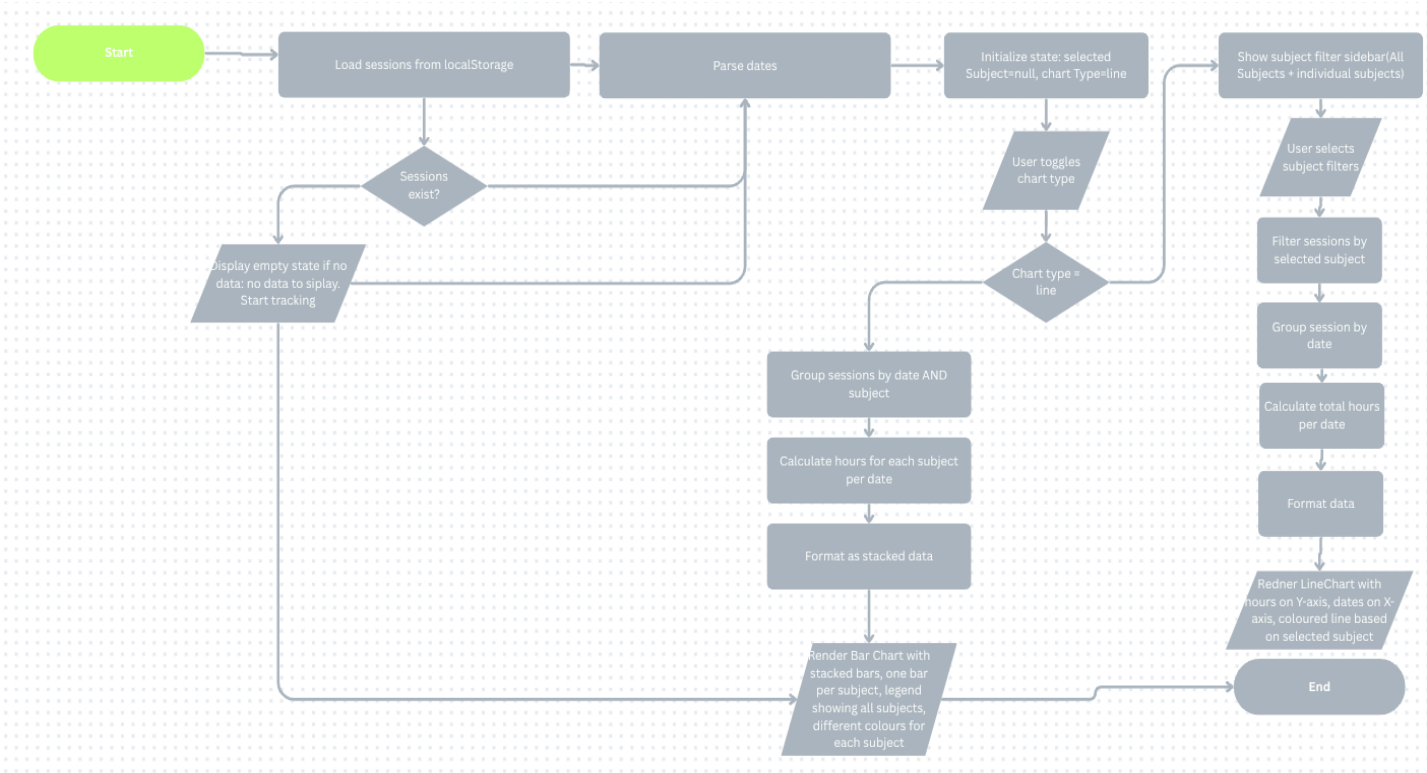

Figure 2: Flowchart that shows the functionalities of the charts

Pseudocode

```
BEGIN calculateTotalStudyTime
    totalTime=1
    FOR each session in StudySessions
        TotalTime=totalTime+session duration
    ENDFOR
    RETURN totalTime
END
```

Figure 3: Pseudocode for calculating total study hours

```
BEGIN LoginUser
    INPUT email, password
    SEND credentials to authentication serivice
    IF credentials are valid THEN
        create user session
        Redirect to dashboard
    ELSE
        display error message
    ENDIF
END
```

Figure 4: Pseudocode showing the authentication process

UI Design

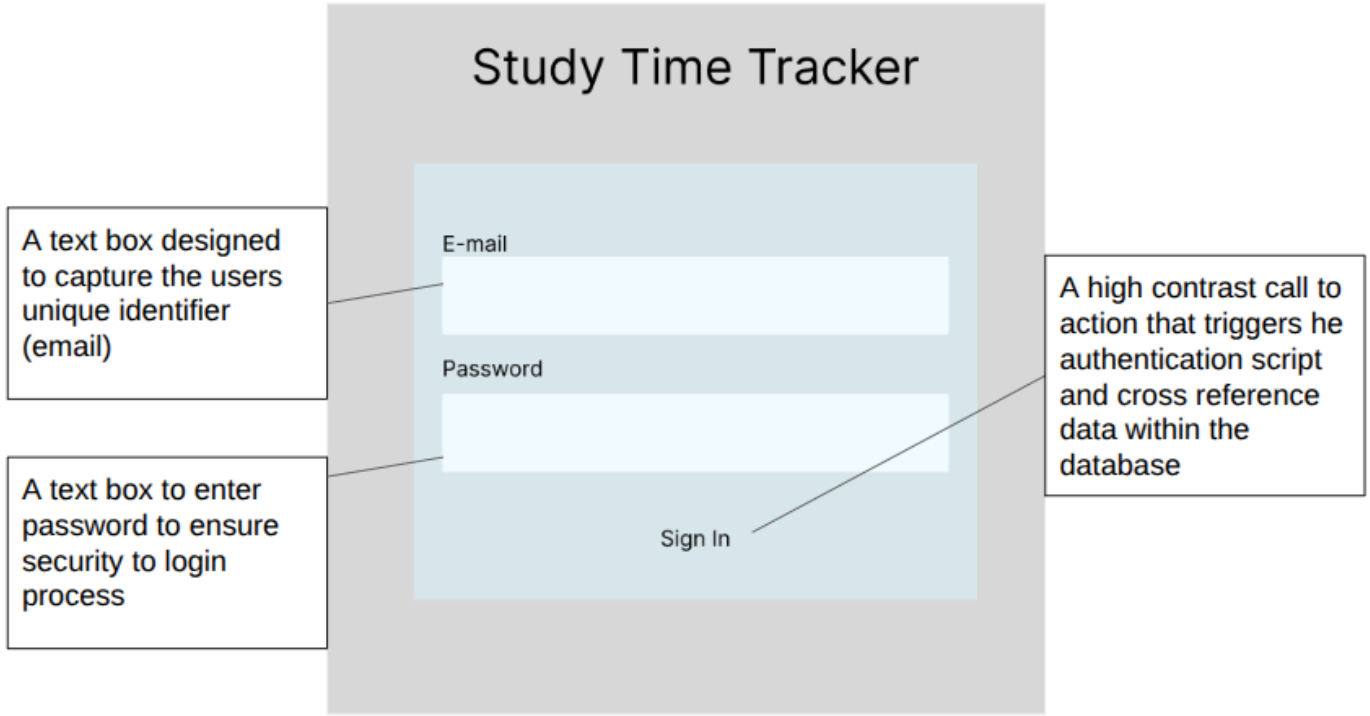

Figure 5: Login page

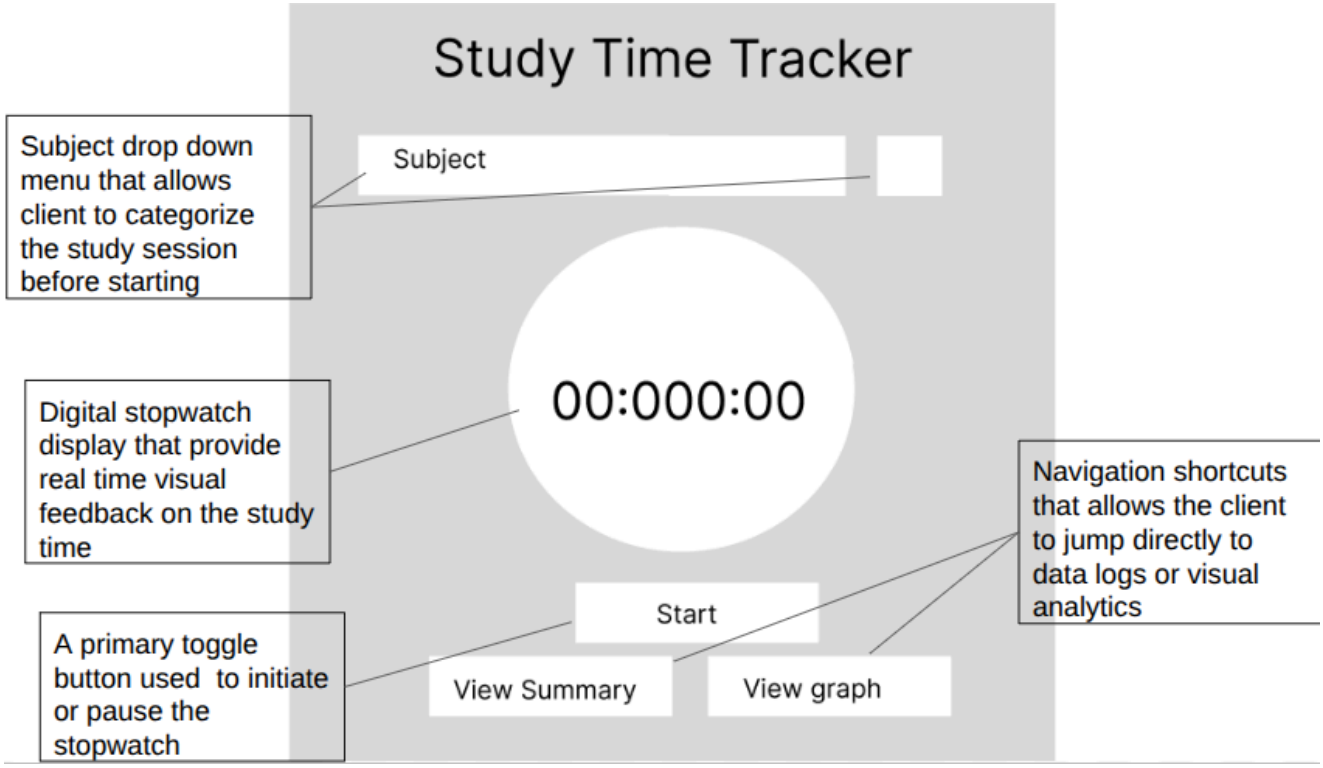

Figure 6: Timer page

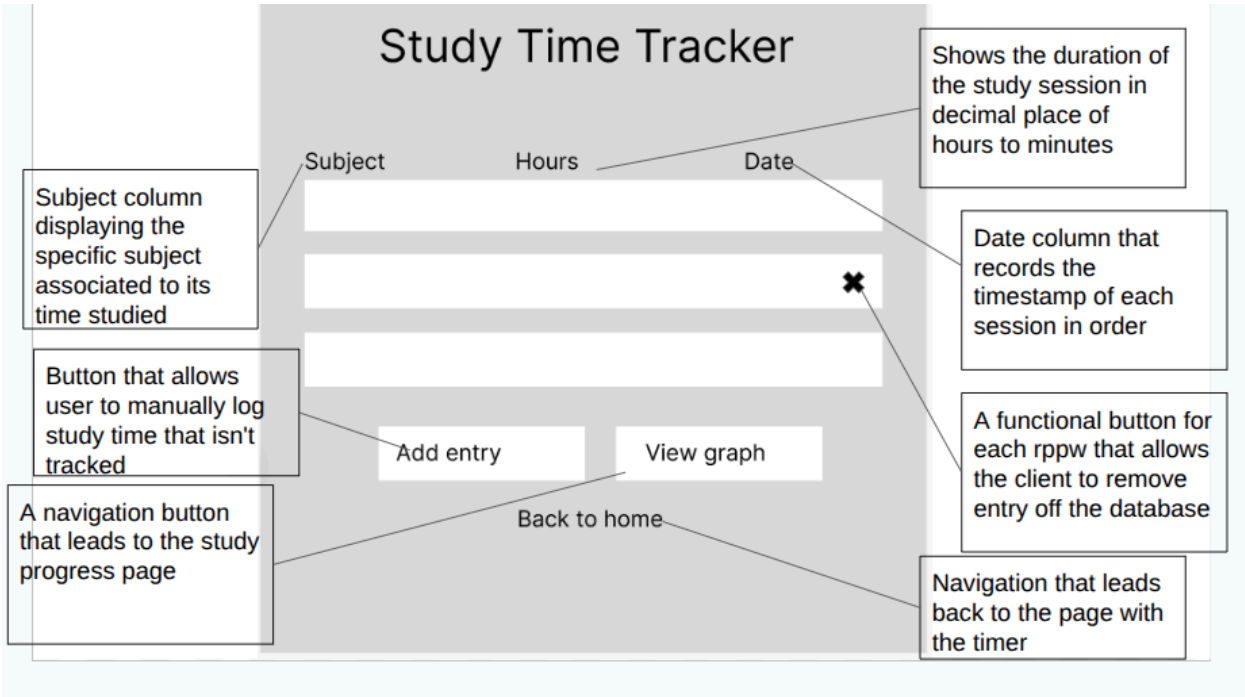

Figure 7: All study record portrayed

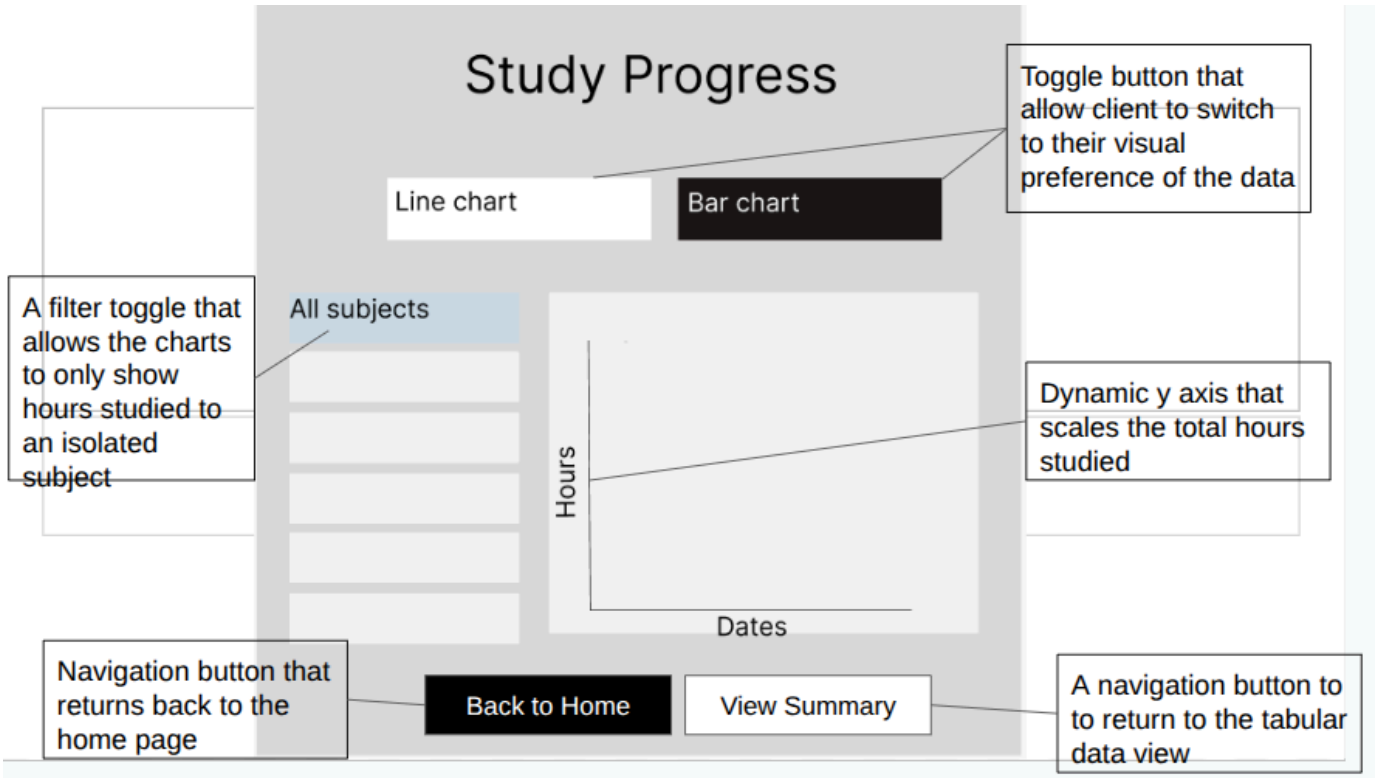

Figure 8: Charts page representation

Test Plan

Test	Success Criterion	Test Description	Test method	Expected Results
1	SC1	Email and password	Input email and password in the login	User is successfully logged in
2	SC1	Log in using invalid credentials	Input invalid email and password in the login	Login is rejected with an7
2	SC1	The timer should be in numerical order and can be paused.	Click the start and pause button.	The timer goes up in numerical order and stops when paused.
3	SC2	Try entering a subject, hours and a date in the input sections.	Subject: English Hours: 2.5 Date: 08/12/2025	All the sections accept the input successfully without any errors.
4	SC2	Add a new study	Subject: Math	The subject

		subject		appears immediately onto the page
5	SC2	Delete a study subject	Subject: Biology	Subject is removed permanently
6	SC12	Click the X on one of the summary entries.	Click on any entry.	The entry that is picked is permanently removed.
7	SC4	Click on the line chat and the bar chart	Place mouse on the line/bar	The same amount of daily hours should show up on both
8	SC3	Run a study session for only 65 seconds	Start and stop timer manually	Recorded duration = 65 seconds
9	SC9	Attempt overlapping sessions	Start new session whilst one is active	System blocks second session

10	SC6	Got to the Study Progress page.	Multiple saved subjects	The chart correctly displays the total studied hours per subject.
11	SC13, SC6 and SC5	Select the subject from the filter dropdown.	Choose: Maths	Chart updates should only show the Math data.
12	SC15 and SC14	Use each navigation button: Enter Data, View Summary, View Graph, Back to Home.		Each button should take you to the correct page without any error.
13	SC6 and SC10	Evaluate UI clarity, readability and the ease of use.		The interface feels intuitive, it’s clear and easy to navigate.
14	SC15 and SC11	Add data, then	any entry	Data will remain

		switch pages and then return.		unchanged across pages (no loss).
15	SC8	Reload the browser		The data should look the same from before and after the reloading.